🎯 Project Summary

Goal:

Compare **SGD vs SVRG vs SAGA** on **L2-regularized logistic regression** using the **Breast Cancer dataset**, focusing on:

- convergence
- runtime
- memory

✅ 1. Problem Setup

Dataset

- Use: sklearn.datasets.load_breast_cancer
- Standardize features
- Convert labels to:

  $y \in \{-1, +1\}$

Objective

$$F(w,b)=\frac{1}{n}\sum_{i=1}^n \log(1+\exp(-y_i(w^\top x_i+b)))+\frac{\lambda}{2}\|w\|_2^2$$

- Use: $\lambda = 10^{-3}$
- Do **not** regularize bias

✅ 2. Implementation Order

Step 1: Data + Objective (YOU)

- load dataset
- normalize features
- implement:
  - loss function
  - per-sample gradient
  - full gradient

Step 2: SGD (YOU)

- random sample per step
- update parameters
- track:
  - loss
  - gradient evaluations
  - runtime

👉 Make sure this works before anything else

Step 3: SVRG (EITHER / BOTH)

- compute full gradient at snapshot
- run inner loop with variance-reduced updates
- set:
  - snapshot frequency ≈ n
- reuse your SGD structure

Step 4: SAGA (WILLIAM)

- store per-sample gradients
- maintain running average

- update using corrected gradient
- update stored gradient after each step

✅ 3. Fair Comparison Setup

Keep SAME for all methods:

- initialization
- dataset split
- preprocessing
- stopping condition
- number of epochs

Tune learning rate per method (small grid search)

✅ 4. Metrics to Track

For each method, record:

- objective value
- number of gradient evaluations
- wall-clock runtime

✅ 5. Core Experiments

Experiment 1 (MAIN)

Compare:

- SGD vs SVRG vs SAGA

Plot:

- loss vs gradient evaluations

- loss vs runtime

Experiment 2

Learning rate sensitivity:

- test multiple step sizes
- analyze stability

Experiment 3 (TRADEOFF)

Create table:

Method	Memory	Compute	Runtime	Final Loss

✅ 6. Tradeoff Analysis (MOST IMPORTANT)

You must explicitly answer:

- Which converges fastest?
- Which is fastest in runtime?
- Which uses most memory?
- When is SAGA worth its memory cost?
- How does SVRG compare in compute vs memory?

✅ 7. Plot Requirements

Make clean plots:

- same scale
- labeled curves
- multiple methods on same graph

Main plots:

- loss vs gradient evaluations
- loss vs runtime

✅ 8. Roles

William

- implement SAGA
- understand SAGA + SVRG theory

You

- implement SGD
- build experiment pipeline
- measure runtime + memory
- generate plots

Both

- debug
- analyze results
- write report

✅ 9. Minimum viable version (if short on time)

- 1 dataset (Breast Cancer)
- logistic regression
- SGD vs SVRG vs SAGA

- 2 plots (eval + runtime)
- 1 tradeoff table
- clear conclusions

👉 This is enough for an A if done cleanly

🚀 Final mindset

> Keep everything simple, consistent, and clean.

> The grade comes from
>
> **clear comparisons + strong insights**